

Lecture 4, Mar 23 2026.

11.1. Sequences.

Def. Let consider $f: \mathbb{N} \rightarrow \mathbb{R}$ then $f(0) = a_0, f(1) = a_1, \dots$,
 $f(n) = a_n \dots$ i.e., the graph of f is called a sequence.

Notation: $\{a_0, a_1, a_2, \dots, a_n\}$ is a sequence which is also denoted by
 $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$.

Example. Sometimes the sequences can be defined by giving a formula for the n th term.

$$\bullet \left\{ \frac{1}{2^n} \right\}_{n=1}^{\infty} = \left\{ \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots, \frac{1}{2^n}, \dots \right\}.$$

$$a_1 = \frac{1}{2}, a_2 = \frac{1}{2^2}, \dots, a_n = \frac{1}{2^n}.$$

$$\bullet \left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty} \quad \text{the formula is} \quad a_n = \frac{n}{n+1}.$$

$$\left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \right\}.$$

$$\bullet \{ \sqrt{3}, \sqrt{4}, \sqrt{5}, \sqrt{6}, \dots \}. \quad a_n = \sqrt{2+n}.$$

$$a_1 = \sqrt{3}, a_2 = \sqrt{4}, a_3 = \sqrt{5},$$

$$\bullet \left\{ (-1)^n \frac{(n+1)}{3^n} \right\}_{n=1}^{\infty} = \left\{ -\frac{2}{3}, \frac{3}{3^2}, -\frac{4}{3^3}, \dots \right\}.$$

Example. Find a formula for the general term a_n of the sequence

$$\left\{ \frac{3}{5}, -\frac{4}{25}, \frac{5}{125}, -\frac{6}{625}, \frac{7}{3125}, \dots \right\}.$$

Soln. we claim that

$$a_n = \frac{(-1)^{n+1}(n+2)}{5^n}$$

The Limit of a Sequence.

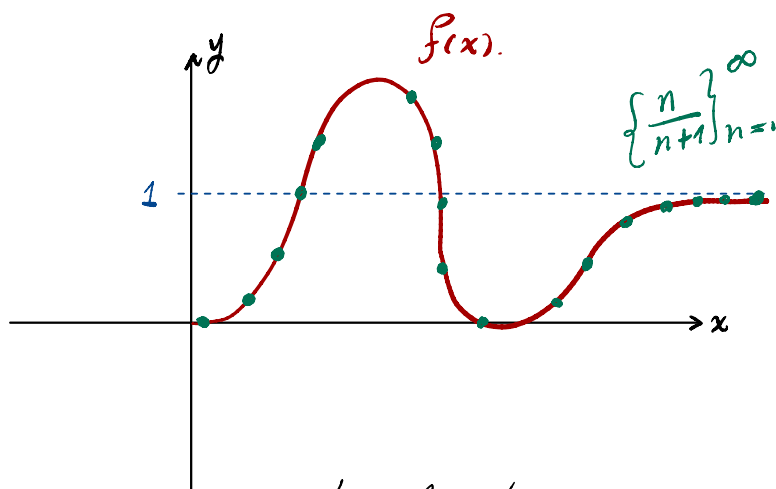
Let consider the sequence

$$\left\{ \frac{n}{n+1} \right\} = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots \right\}.$$

Notice that $f(x) = \frac{x}{x+1}$ $\text{Dom } f = \mathbb{R} - \{-1\}$ then

$$\lim_{x \rightarrow \infty} \frac{x}{x+1} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} (1) = 1.$$

So,



then $\left\{ \frac{n}{n+1} \right\}$ is convergent to 1.

Def. A sequence $\{a_n\}$ has limit L and we write
$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \text{ as } n \rightarrow \infty.$$

if we make the terms a_n as close to L as we like by taking n sufficiently large.

Properties of convergent sequences.

Thm If $\lim_{x \rightarrow \infty} f(x) = L$ and $f(n) = a_n$ when $n \in \mathbb{N}$, then
$$\lim_{n \rightarrow \infty} a_n = L.$$

Limits law for sequences. if $\{a_n\}$ and $\{b_n\}$ are convergent sequences.

0). $\lim_{n \rightarrow \infty} \frac{1}{n^r} = 0$ if $r > 0$.

1). $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$

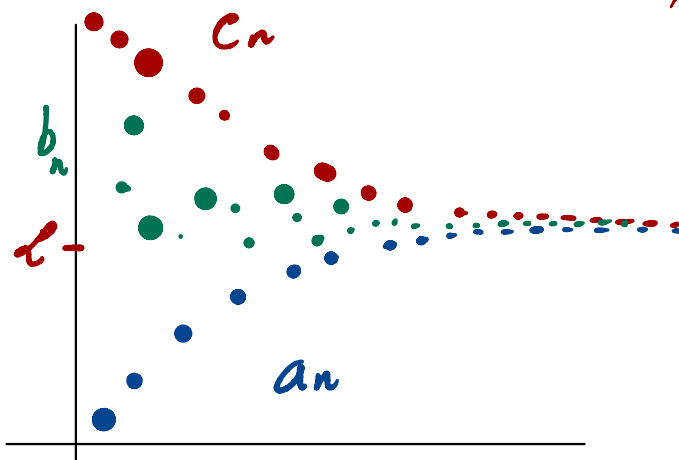
$$2). \lim_{n \rightarrow \infty} (c \cdot a_n) = c \cdot \lim_{n \rightarrow \infty} a_n.$$

$$3). \lim_{n \rightarrow \infty} (a_n \cdot b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right).$$

$$4). \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} \quad \text{if } \lim_{n \rightarrow \infty} b_n \neq 0.$$

$$5). \lim_{n \rightarrow \infty} a_n^p = \left[\lim_{n \rightarrow \infty} a_n \right]^p \quad \text{if } p > 0 \text{ and } a_n > 0.$$

$$6). (\text{Squeeze thm}) \quad \text{if } a_n \leq b_n \leq c_n \text{ for } n \geq n_0 \text{ and } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L.$$



$$\text{then } \lim_{n \rightarrow \infty} b_n = L.$$

$$7). \text{ If } \lim_{n \rightarrow \infty} |a_n| = 0 \text{ then } \lim_{n \rightarrow \infty} a_n = 0.$$

Examples: a). Is the sequence $a_n = \frac{n}{\sqrt{10+n}}$ convergent or divergent?

This is divergent b/c.

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt{10+n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{10/n^2 + 1/n}} = \infty.$$

$$b). \text{ Compute } \lim_{n \rightarrow \infty} \frac{\ln n}{n}.$$

Slm. We can't apply L'Hospital's rule directly b/c it applies not to the sequences but to function of a real variable.

Let $f(x) = \frac{\ln x}{x}$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

So, by thm $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$.

Example. Determine whether the sequence $a_n = (-1)^n$ is convergent or divergent.

Sln. $\{-1, 1, -1, 1, -1, 1, \dots\}$.

Since the terms oscillate between 1 and -1 infinitely often, a_n does not approach any number.

Example. Evaluate $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$ if it exists.

Sln. (ST). $-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}$ for all n .

$\downarrow$ as $n \rightarrow \infty$ $\downarrow$ as $n \rightarrow \infty$
 0 0

then, by ST $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$.

(ST ST). $\lim_{n \rightarrow \infty} |a_n| = \lim_{n \rightarrow \infty} |(-1)^n/n| = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$. By thm

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$$

Thm. If $\lim_{n \rightarrow \infty} a_n = L$ and the function f is continuous at L , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L).$$

Example. Find $\lim_{n \rightarrow \infty} \sin \frac{\pi}{n}$

Sln. Sine function is continuous at 0 and $\lim_{n \rightarrow \infty} \frac{\pi}{n} = 0$ so,
$$\lim_{n \rightarrow \infty} \sin \frac{\pi}{n} = \sin(0) = 0.$$

Example. Discuss the convergence of the sequence $a_n = \frac{n!}{n^n}$ where $n! = 1 \cdot 2 \cdot 3 \cdots n$.

Sln.

$$0 \leq a_n = \frac{n!}{n^n} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdots (n-1) \cdot n}{\underbrace{n \cdot n \cdot n \cdots n}_{n\text{-times}}} = \frac{1}{n} \left(\frac{2 \cdot 3 \cdot 4 \cdots n}{n \cdot n \cdots n} \right) \leq \frac{1}{n}$$

Since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ By S-T we have

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n}.$$

Problem For what values of r is the sequence $\{r^n\}$ convergent?